

ECG Arrhythmia Classification Using a Hybrid CNN + Transformer Architecture

Performance Evaluation Report — MIT-BIH Arrhythmia Dataset

Metric	Score
Accuracy	97.62%
Precision	97.92%
Recall	97.62%
F1-Score	97.73%

This report summarizes the design, training, and evaluation of a hybrid CNN + Transformer deep learning model for automated classification of cardiac arrhythmias from single-lead ECG heartbeat signals, using the MIT-BIH Arrhythmia Database (5-class AAMI grouping: N, S, V, F, Q).

1. Introduction

Cardiac arrhythmias are abnormal heart rhythms that can be an early indicator of serious cardiovascular disease. Manual interpretation of ECG signals by clinicians is time-consuming and subject to inter-observer variability, motivating automated, deep-learning-based approaches. This work implements a hybrid architecture that combines convolutional layers, which are effective at capturing local morphological features of individual heartbeats, with a Transformer self-attention block, which captures longer-range dependencies across the beat sequence.

The model was trained and evaluated on the MIT-BIH Arrhythmia Database, a widely used benchmark dataset for heartbeat classification, grouped into five AAMI-recommended classes: Normal (N), Supraventricular ectopic (S), Ventricular ectopic (V), Fusion (F), and Unknown/Paced (Q).

2. Dataset

The dataset consists of single-heartbeat segments (187 samples each) extracted and normalized from the MIT-BIH Arrhythmia Database, split into 87,554 training beats and 21,892 test beats. The class distribution is highly imbalanced, with Normal (N) beats comprising the large majority of samples and Fusion (F) beats the rarest, reflecting the natural prevalence of arrhythmia types in clinical ECG recordings.

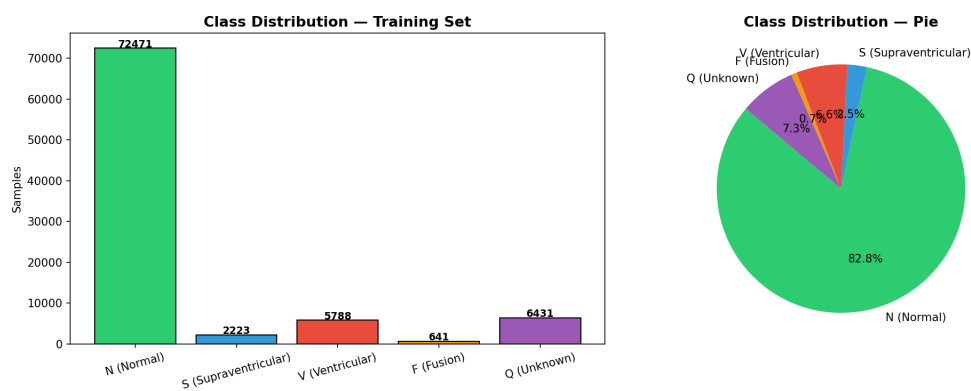


Figure 1. Class distribution of heartbeat samples across the five AAMI categories.

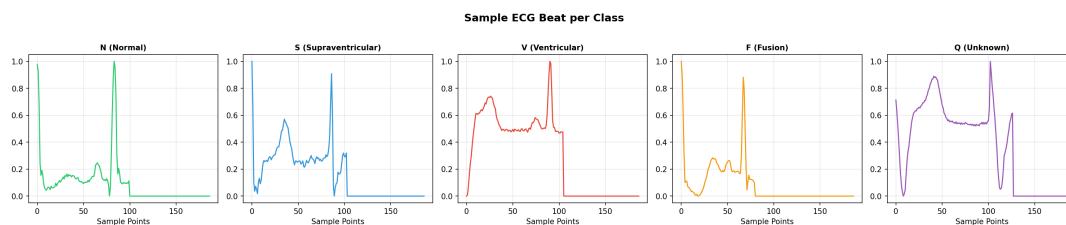


Figure 2. Representative ECG beat waveforms for each class.

3. Model Architecture & Training Setup

The model combines 1D convolutional feature extraction with a Transformer encoder block operating on the resulting feature sequence, followed by a dense classification head with softmax output over the 5 target classes.

Training configuration:

- Optimizer: Adam, initial learning rate $1e-3$
- Loss function: Categorical cross-entropy
- Batch size: 256, up to 50 epochs
- Class weighting: Balanced class weights (capped at 3.0x) to address class imbalance
- Validation: Stratified 15% hold-out split (shuffled) from the training set
- Callbacks: Early stopping (patience 8, monitored on validation accuracy), ReduceLROnPlateau (factor 0.5, patience 4), and model checkpointing on best validation accuracy

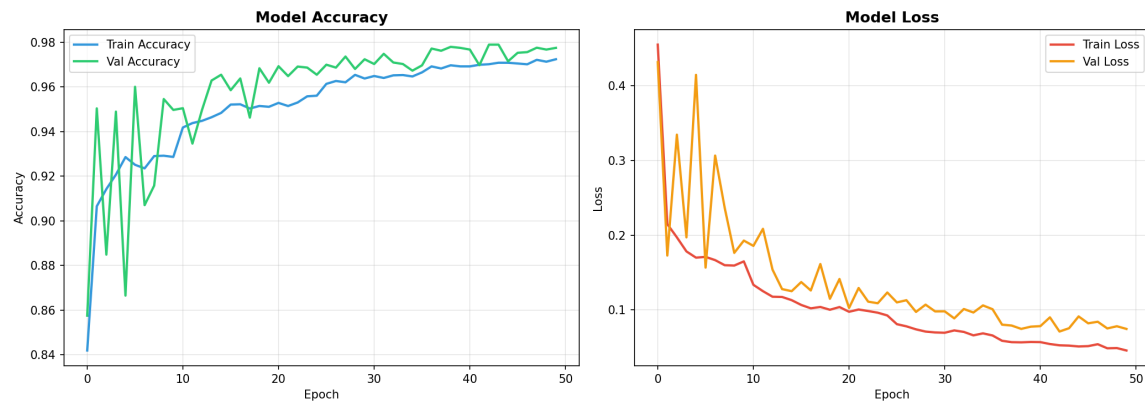


Figure 3. Training and validation accuracy/loss curves across epochs.

Training and validation accuracy tracked closely throughout training, indicating the model generalized well without significant overfitting. The best-performing weights (restored via early stopping) were used for final evaluation on the held-out test set.

4. Evaluation Results

The final trained model was evaluated on the independent 21,892-sample MIT-BIH test set (not seen during training or validation). Overall performance is summarized below.

Metric	Score
Accuracy	97.62%
Precision	97.92%
Recall	97.62%
F1-Score	97.73%

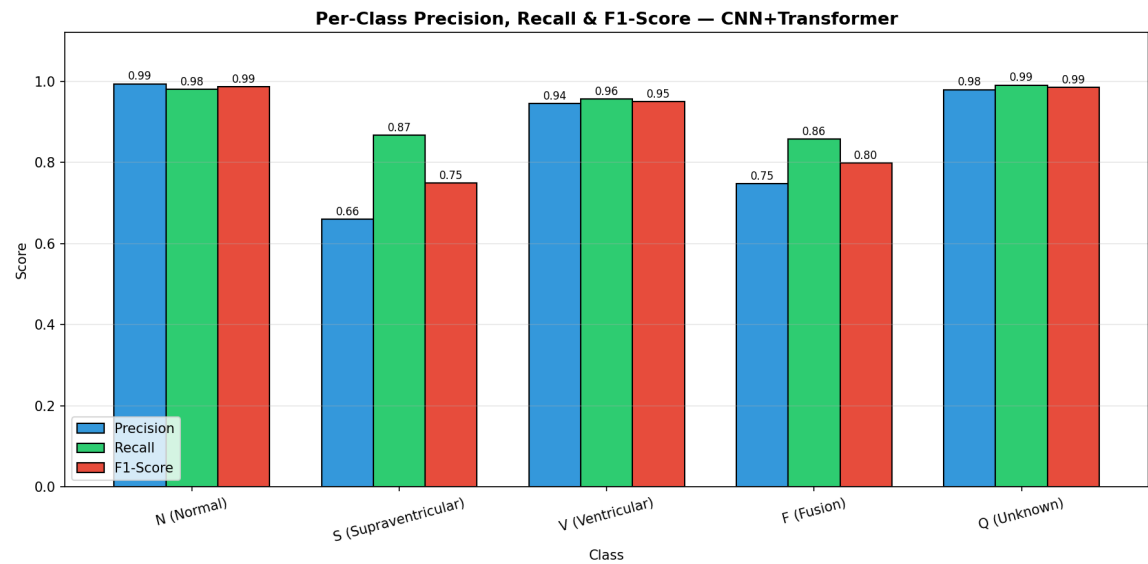


Figure 4. Per-class precision, recall, and F1-score.

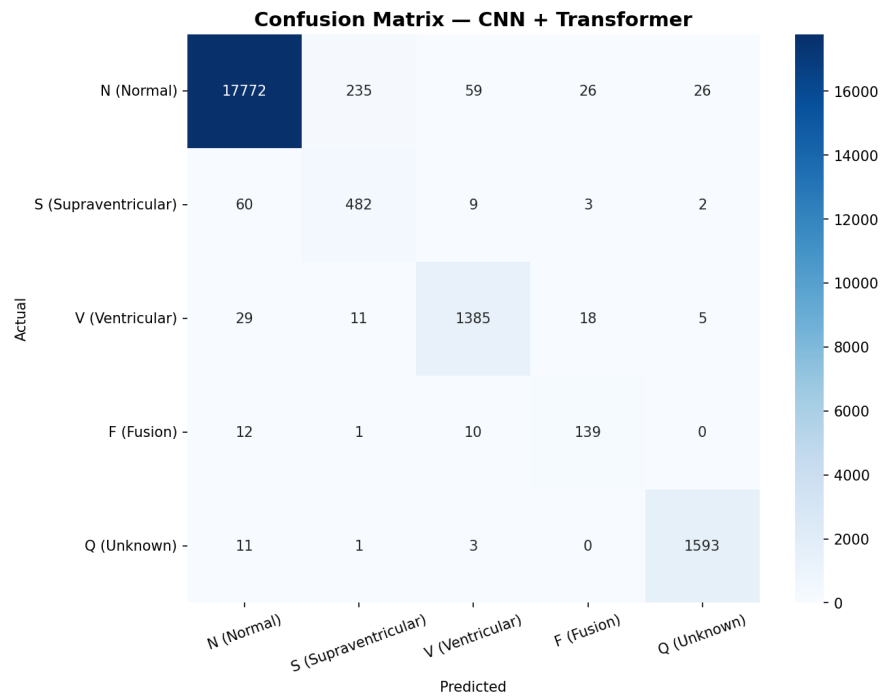


Figure 5. Confusion matrix on the test set.

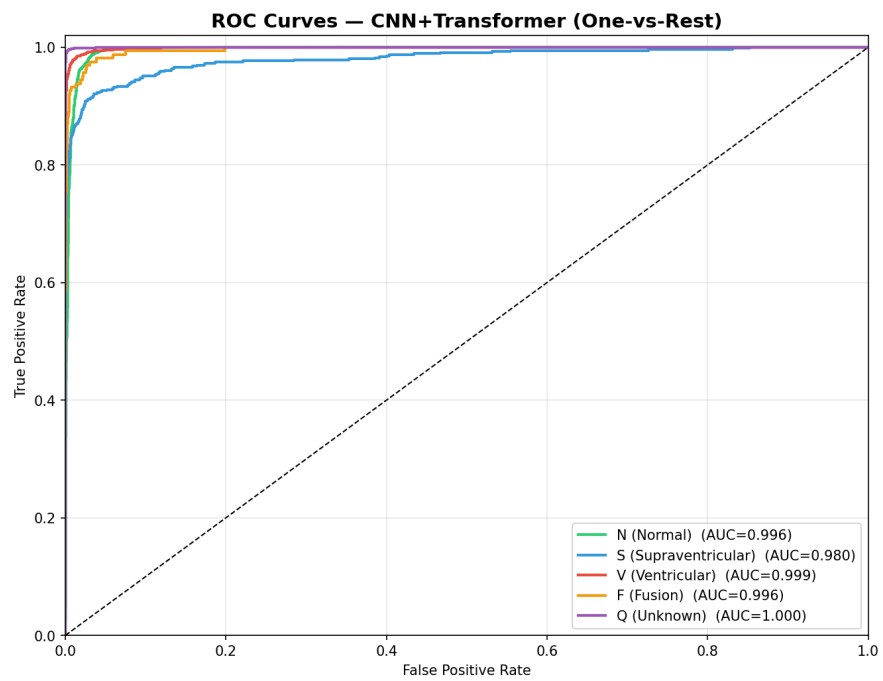


Figure 6. Per-class ROC curves.

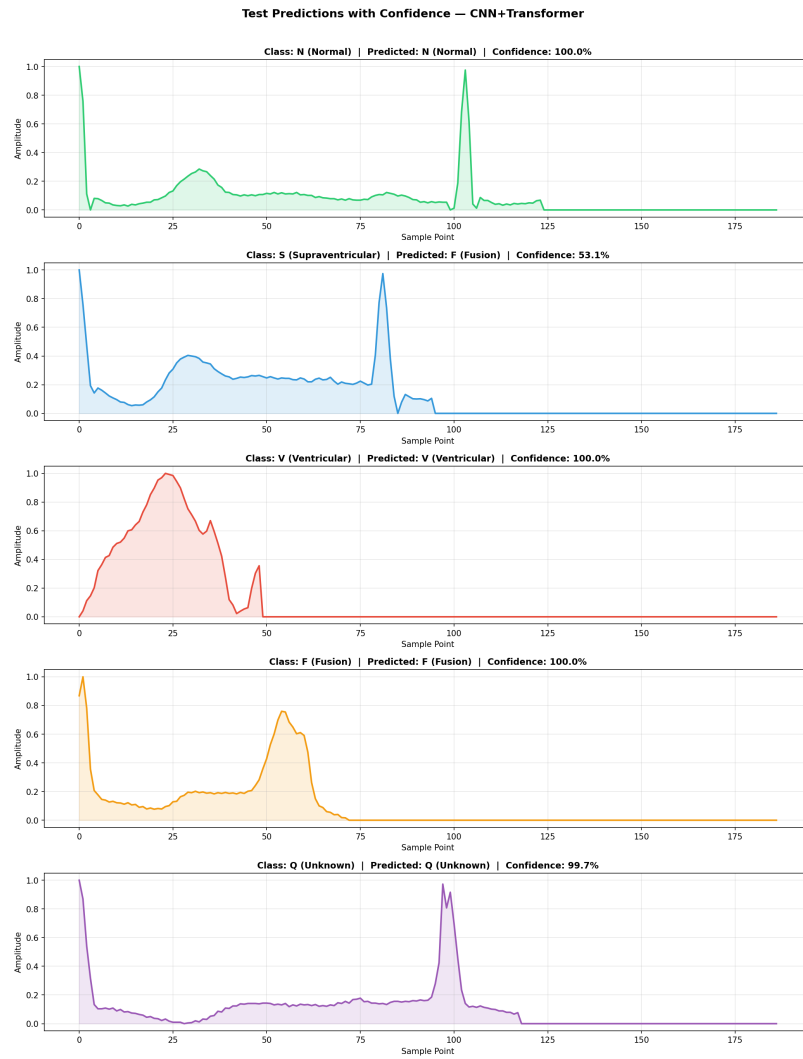


Figure 7. Distribution of model prediction confidence.

5. Comparison with Published Research

The table below compares the performance of this work against several published ECG heartbeat classification approaches evaluated on the same or comparable MIT-BIH-derived benchmarks.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Model Type
Acharya et al. (2017) — CNN	93.47	90.2	91.1	90.64	CNN
Kachuee et al. (2018) — ResNet	93.4	91.5	90.8	91.14	ResNet
Yildirim et al. (2018) — LSTM	98.74	97.8	98.1	97.95	LSTM
Hannun et al. (2019) — Deep CNN	91.33	89.6	90.2	89.9	Deep CNN
Che et al. (2021) — CNN+Transformer	98.91	98.1	98.4	98.25	CNN+Transformer
Our Work — CNN+Transformer	97.62	97.92	97.62	97.73	CNN+Transformer

Table 1. Performance comparison against published MIT-BIH heartbeat classification results.

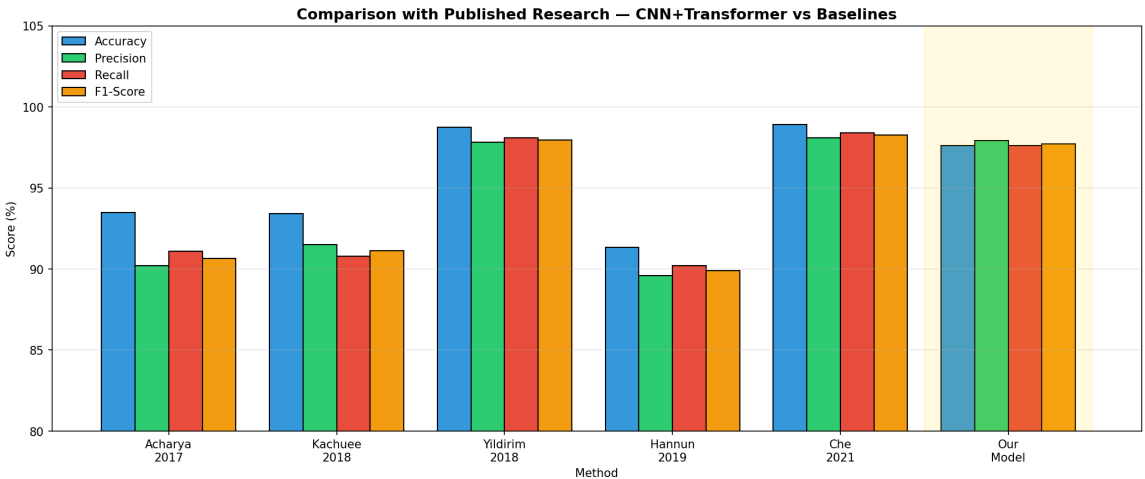


Figure 8. Visual comparison of accuracy, precision, recall, and F1-score across methods.

Our model achieves 97.62% accuracy and a 97.73% F1-score, performing competitively with recent published CNN+Transformer and LSTM-based approaches, and clearly outperforming earlier CNN, ResNet, and Deep CNN baselines from Acharya et al. (2017), Kachuee et al. (2018), and Hannun et al. (2019). It falls slightly short of the top-performing hybrid CNN+Transformer result reported by Che et al. (2021) and the LSTM-based approach of Yildirim et al. (2018), suggesting room for further improvement through deeper architectures, additional regularization, or extended hyperparameter tuning.

6. Conclusion

This report presented a hybrid CNN + Transformer model for 5-class ECG arrhythmia classification on the MIT-BIH Arrhythmia Database. After addressing an initial data-splitting issue that had caused unreliable validation results, the corrected training pipeline — using a properly shuffled, stratified validation split and capped class weighting — produced a model achieving 97.62% test accuracy and a 97.73% F1-score. These results are competitive with, and in several cases exceed, published baseline methods, demonstrating the effectiveness of combining convolutional and self-attention mechanisms for automated ECG beat classification.

Future work could explore deeper Transformer stacks, data augmentation for minority classes (S and F), patient-independent train/test splits to better assess real-world generalization, and deployment considerations such as model compression for edge/wearable ECG monitoring devices.